

Chacha Chen

CONTACT Mobile: +1 (814)-699-2243
INFORMATION Email: chachachen03@gmail.com
Website: <https://chacha-chen.github.io>

RESEARCH Human-centered AI, at the intersection of machine learning, computational social science,
KEYWORDS and human-computer interaction.

EDUCATION **The University of Chicago** Aug. 2021 - current
Department of Computer Science
PhD in Computer Science
• **Advisor:** Chenhao Tan

The Pennsylvania State University Aug. 2019 - Dec. 2020
College of Information Sciences and Technologies
Master in Informatics
• **Advisor:** Ting-Hao (Kenneth) Huang, Zhenhui (Jessie) Li

Shanghai Jiao Tong University Sep. 2015 – Jun. 2019
Department of Computer Science
Bachelor of Science in Computer Science
• **Advisor:** Weinan Zhang, Zhenhui (Jessie) Li

RESEARCH & **IQVIA, Analytics Center of Excellence, Boston**
WORK Machine Learning Research Intern May. 2020 – Aug. 2020
EXPERIENCE • Gaussian Process for health risk prediction: Designed Bayesian machine learning models for informative health risk prediction. Different from previous methods, our proposed model is able to give uncertainty-based rich prediction. Moreover, we leveraged multi-sourced data with different modalities.
• Supervised by Cao (Danica) Xiao (IQVIA), Prof. Fenglong Ma (PSU) and Prof. Jimeng Sun (UIUC).

PUBLICATIONS **Conference Papers**
1. Guanjie Zheng, Chang Liu, Hua Wei, Porter Jenkins, **Chacha Chen**, Tao Wen, Zhenhui Li. Knowledge-based Residual Learning. In Proceedings of the The 30th International Joint Conference on Artificial Intelligence (**IJCAI 2021**), Aug 2021.
2. **Chacha Chen**, Junjie Liang, Fenglong Ma, Lucas Glass, Jimeng Sun, Cao Xiao, UNITE: Uncertainty-based Health Risk Prediction Leveraging Multi-sourced Data, In Proceedings of The Web Conference 2021 (**WWW 2021**), Ljubljana, Slovenia, April 2021.
3. Guanjie Zheng, Chang Liu, Hua Wei, **Chacha Chen**, Zhenhui Li, Rebuilding City-Wide Traffic from Multi-Source Data. In Proceedings of 37th IEEE International Conference on Data Engineering (**ICDE 2021**), Chania, Greece, April 2021.
4. Hua Wei, **Chacha Chen**, Chang Liu, Guanjie Zheng and Zhenhui Li, ImIn-GAIL: Learning to Simulate with Imitation-Interpolation on Sparse Trajectory Data. In Proceedings of European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (**ECML-PKDD 2020**), Ghent, Belgium, Sep 2020.
Best Applied Data Science Paper Award
5. **Chacha Chen**, Hua Wei, Nan Xu, Guanjie Zheng, Ming Yang, Yuanhao Xiong, Kai Xu and Zhenhui Li, Toward A Thousand Lights: Decentralized Deep Reinforcement

Learning for Large-Scale Traffic Signal Control. In Proceedings of the Thirty-Fourth AAAI Conference on Artificial Intelligence (**AAAI 2020**), New York, United States, February 2020. (Acceptance rate: 1591/7737=20.6%)

6. Hua Wei, Nan Xu, Huichu Zhang, Guanjie Zheng, Xinshi Zang, **Chacha Chen**, Weinan Zhang, Yanmin Zhu, Kai Xu and Zhenhui Li, CoLight: Learning Network-level Cooperation for Traffic Signal Control. In Proceedings of the 2019 ACM on Conference on Information and Knowledge Management (**CIKM 2019**), Beijing, China, November 2019. (Acceptance rate: 200/1030=19.4%)
7. Hua Wei, **Chacha Chen**, Guanjie Zheng, Kan Wu, Vikash V. Gayah, Kai Xu and Zhenhui Li, PressLight: Learning Max Pressure Control to Coordinate Traffic Signals in Arterial Network. In Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD 2019**), Anchorage, AK, USA, Aug 2019. (Acceptance rate: 170/1200=14.2%)

Workshop Papers

1. Tongan Cai*, **Chacha Chen***, Ting-Hao K. Huang, Frank E. Ritter, What Makes A Good Reference Manager? A Quantitative Analysis of Bibliography Management Applications, Asian CHI Symposium 2021 (**AsianCHI2021**).
Best Paper Award
2. Jiaqi Wang, Hua Shen, **Chacha Chen**, Frank E. Ritter, Are Learners Satisfied with their MOOC Experiences? Assessing and Improving Online Learners' Interactions, Asian CHI Symposium 2021 (**AsianCHI2021**).
3. **Chacha Chen**, Guanjie Zheng, Hua Wei, Zhenhui Li, Physics-informed Generative Adversarial Networks. In Proceedings of the NeurIPS Workshop on Interpretable Inductive Biases and Physically Structured Learning (**IIBPSL, NeurIPS 2020**).
4. **Chacha Chen**, Chieh-Yang Huang, Yaqi Hou, Yang Shi, Enyan Dai, Jiaqi Wang, Joint Event Multi-task Learning for Slot Filling in Noisy Text. In Proceedings of the Workshop on Noisy User-generated Text (**W-NUT, EMNLP 2020**).
5. Hua Wei, **Chacha Chen**, Guanjie Zheng, Kan Wu, Vikash V. Gayah, Kai Xu and Zhenhui Li, Deep Reinforcement Learning for Traffic Signal Control along Arterials. In Proceedings of the Workshop on Deep Reinforcement Learning for Knowledge Discovery (**DRL4KDD, KDD 2019**).

COMMUNITY ACTIVITIES & SERVICE	Conference Program Committee Member/Reviewer • AAAI 2021: The Thirty-Fifth AAAI Conference on Artificial Intelligence (AAAI-21) • W-NUT: The 6th Workshop on Noisy User-generated Text with EMNLP 2020 • NeurIPS Workshop on Interpretable Inductive Biases and Physically Structured Learning	
TEACHING EXPERIENCE	Pennsylvania State University Teaching Assistant IST 261 (Java Course): Application Development Design Studio I Prof. Kenneth Huang	Fall 2020
AWARDS	AAAI-20 VISA Research Scholarship AAAI-20 Student Travel Grant Top 25 in the finalist, <i>KDD Cup 2020 LDR Competition</i> Distinguished Bachelor Thesis, <i>Shanghai Jiao Tong University</i> Academic Excellence Scholarship, <i>Shanghai Jiao Tong University</i>	Feb. 2020 Feb. 2020 Aug. 2020 Jun. 2019 Sep. 2016
RELEVANT SKILLS	Technical: Python, C/C++, MySQL, Java, Matlab, R, HTML/CSS Languages: TOEFL (112, Reading 27, Listening 30, Speaking 25, Writing 30)	